

PIXE Analysis on Maya Blue in Prehispanic and Colonial Mural Paintings

Group_Project_8_Presentation

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Overview of Project

Questions to Answer

There are 4 main questions that we set out to answer during this project

1. Are there any surprising observations in the data?
2. Can we assume that the data follows a multivariate normal distribution?
3. Can we find groups in this data?
4. Can we recreate figure 2 from the original study?

Background

Original Data



Figure 2. from the original Paper

Data Transformation

Descriptive Statistics of Elemental Concentrations								
Log10 Transformation								
Element	Mean	Standard D.	Median	Minimum	1st Quartile	Inter-Q Range	3rd Quartile	Maximum
11-Na*	-0.44	0.28	-0.48	-0.96	-0.65	0.35	-0.30	0.23
12-Mg*	0.33	0.35	0.24	-0.42	0.13	0.37	0.51	0.97
13-Al*	0.50	0.19	0.52	0.01	0.41	0.24	0.65	0.87
14-Si*	1.09	0.22	1.06	0.66	0.93	0.32	1.25	1.49
16-S*	-0.23	0.61	-0.33	-1.22	-0.73	0.81	0.08	1.11
17-Cl*	-1.14	0.63	-1.15	-2.30	-1.41	0.45	-0.96	0.13
19-K*	-0.47	0.35	-0.49	-1.22	-0.72	0.51	-0.21	0.23
20-Ca*	1.50	0.26	1.58	0.46	1.47	0.19	1.66	1.76
26-Fe*	-0.02	0.25	-0.03	-0.42	-0.21	0.38	0.16	0.71
15-P	2.53	0.78	1.93	1.93	1.93	1.11	3.04	5.00
22-Ti	3.18	0.28	3.17	2.72	2.93	0.42	3.35	3.70
25-Mn	2.67	0.55	2.49	2.08	2.34	0.43	2.77	3.96
28-Ni	1.23	0.27	1.23	0.48	1.08	0.30	1.38	2.19
29-Cu	0.95	0.38	1.00	0.00	0.70	0.52	1.22	1.60
30-Zn	1.38	0.37	1.32	0.78	1.11	0.59	1.70	2.21
38-Sr	2.74	0.45	2.70	1.95	2.45	0.60	3.05	3.93
82-Pb	0.96	0.61	0.95	0.00	0.60	0.69	1.29	2.18

Figure 1: Log Transformed Data

Methodology

Quantitative Description

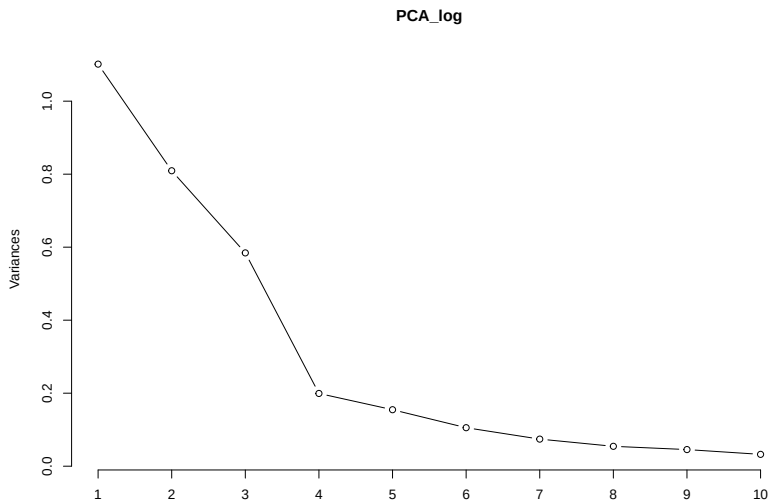
Qualitative Description

Are There Surprising Observations?

Pairs Plot of Mahalanobis Distances and Contour Densitys.

Can we assume that the data follows a multivariate normal distribution?

Can we find groups in this data?



Discussion

The analysis of elemental concentrations within Maya blue samples provides evidence of structure, although the groups are not completely distinct. Taking a log transformation of the data helped to account for large differences in concentration levels between the elements, which is consistent with the approach used in the original study. However, the data cannot be assumed to follow a multivariate normal distribution, as it fails Mardia's skewness test, despite the kurtosis result showing consistency with normality. Several of the individual elements also showed evidence of non-normality; these elements were calcium, phosphorus, manganese and nickel. This should be taken into consideration when interpreting the multivariate results.

The data also showed some surprising observations. TAJ1_4 had the biggest Mahalanobis distance and was classified as surprising, while Tot1_1, TAJ1_1, Jiu1_1 and Cec1_3 were all classified as somewhat surprising. The clustering analysis provides evidence of some meaningful groups in the data, with the CEC samples forming a distinct group. This is consistent with the authors' observation

Conclusion

Overall, the results show some structure in the Maya blue samples, but the groups are not completely distinct. The CEC samples are clearly separated, while the Prehispanic and Colonial samples overlap. Figure 2 supports the authors' conclusion that the zones represent broad patterns rather than clearly defined groups.

References